SHIVANSHU TRIPATHI *Curriculum Vitae/August 2026*

CONTACT INFORMATION

1144 W Blaine Street
Riverside, CA 92507
Google Scholar

(951) 610-7072 (Mob.)
strip008@ucr.edu
shivanshutripathi11@gmail.com

RESEARCH INTERESTS

Agentic AI, physics-informed ML, foundation models, scientific ML, optimization, smart grids.

EDUCATION

University of California, Riverside, USA

Ph.D. + M.S., Electrical and Computer Engineering **Sept 2022 – March 2027**

- Dissertation: *Designing tool-augmented LLMs for power systems.*
- Advisor: Prof. Maziar Raissi, Prof. Hamed Mohsenian-Rad

Indian Institute of Technology Jodhpur, India

M.Tech., Electrical Engineering (Cyber Physical Systems) **August 2020 – April 2022**

- Thesis: *Distributed convex optimization in multi-agent systems*
- Advisor: Prof. Anoop Jain GPA: 8.95/10

National Institute of Technology Srinagar, J&K, India

B.Tech., Electrical Engineering **August 2016 – July 2020**

- Advisors: Prof. Tabish Nazir Mir and Prof. A. H. Bhat GPA: 8.744/10

RESEARCH AND WORK EXPERIENCE

National Laboratory of Rockies (NLR)

Graduate Research Intern **June 2026 – September 2026**

- Building foundation models and AI agents for power systems.
- Advisor: Fei Ding

Pacific Northwest National Laboratory (PNNL)

Graduate Research Intern **November 2025 – March 2026**

- Developed AI agents for power systems.

Department of Telecommunications, Government of India

Research Associate **May 2022 – August 2022**

PUBLICATIONS

Journal Publications

[J1] Shivanshu Tripathi, A. Al Makdah, and F. Pasqualetti, "Online Optimization with Unknown Time-varying Parameters," *IEEE Control Systems Letters, L-CSS, 2025.* Link

[J2] Shivanshu Tripathi, A. Jain, and A. K. Behera, "A Partially Distributed Fixed-Time Economic Dispatch Algorithm with Kron's Modeled Power Transmission Losses," *IEEE Transactions on Control of Network Systems*, June 2023. Link

[J3] Shivanshu Tripathi and M. Raissi, "Online Optimization with Unknown Time-varying Parameters using Noisy Gradient Measurements," submitted to *IEEE Control Systems Letters, L-CSS, 2026.* Link

[J4] S. Shekhar, Shivanshu Tripathi and others "Critical Review of Artificial Intelligence and Machine Learning Methods for Data-Center Load Forecasting" Energies, 2026 Link

Conference Papers

[C1] Shivanshu Tripathi, H. Mohsenian-Rad, and M. Raissi, "VeraGrid-Agent: Tool-Augmented LLMs for Distribution Optimal Power Flow at the Grid Edge," *IEEE PES Grid Edge, 2026.* Link

[C2] K. Bhaumik, Shivanshu Tripathi, and M. Raissi, "MathAgent: A SciPy-Augmented Benchmark for Agentic Scientific Computation," *MLSys 2026.*

[C3] Shivanshu Tripathi, M. Raissi, and H. Mohsenian-Rad, "Data-Efficient Physics-Informed Learning for Synchro-Waveform Analytics of Grid-Integrated Inverter-Based Resources," *IEEE International Conf. on Harmonics and Quality of Power, 2026.* Link

[C4] Shivanshu Tripathi, M. Raissi, and H. Mohsenian-Rad, "Spectrogram-Based Joint Detection, Localization, and Classification of Events in Continuously Recorded IBR Waveforms," *IEEE PES Grid Edge, 2026.* Link

[C5] D. Gadginmath, Shivanshu Tripathi, and F. Pasqualetti, "Fusing Multiple Algorithms for Heterogeneous Online Learning," *American Control Conference*, 2025. Link

[C6] Shivanshu Tripathi, R. Akbarian Bafghi, and M. Raissi, "Test-Driven Agentic Framework for Reliable Robot Controller," *International Conference on Control, Automation, and Systems, ICCAS, 2026.* Link

[C7] Shivanshu Tripathi, M. Raissi "Exact Distributed Attitude Alignment on $SO3$ for Robot Teams with Conflicting Local Objectives," *American Control Conference*, 2027.

Honors and Awards

- Dean's Distinguished Fellowship, University of California, Riverside.
- Gold medal for outstanding student in academics, IIT Jodhpur.
- Best B.Tech. undergraduate thesis in the department, NIT Srinagar.
- Student Travel Award 2025, 64th IEEE Conference on Decision and Control, Rio, Brazil.
- Student Travel Award 2025, 22nd IEEE International Conference on Harmonics and Quality of Power, Dresden, Germany.

Talks and Presentations

- NSF presentation on physics-informed learning, Rutgers University, March 17, 2026.
- NC4 talk, Army Research Laboratory (ARL), September 18, 2025.
- NC4 work, Army Research Laboratory (ARL), September 26, 2024.
- Quarterly project updates to Army Research Labs (funding agency).

Professional Service

Reviewer for:

- **Power systems:** IEEE Transactions on Smart Grid, IEEE Transactions on Dependable and Secure Computing (TDSC), IEEE PES Grid Edge
- **Control theory:** IEEE Control Systems Letters (L-CSS), IEEE Conference on Decision and Control (CDC), American Control Conference (ACC), IEEE Conference on Control Technology and Applications (CCTA), IEEE European Control Conference (ECC), IEEE Transactions on Automatic Control (TAC)

Teaching Experience

- Teaching Assistant, Control Lab (EEL3040 and EEL324).
- Teaching Assistant, Power Engineering Lab and Electrical Machines Lab (EEL3001).

Computer Skills

Programming: Python, C++, MATLAB
Deep learning libraries: PyTorch, TensorFlow
Optimization solvers: CPLEX, OSQP
Software/simulators: PSCAD, OpenDSS, AutoCAD, NS3

Key Coursework

Machine learning: Deep learning, machine learning, computational learning.
Control systems: Linear control theory, non-linear control theory, network dynamical systems, optimal control, data-driven control.
Optimization: Convex optimization.
Power systems: Smart grid and distributed generation, power systems, high voltage transmission systems, power electronics, electric machines.

Mentoring

Kishor Kumar Bhaumik
Ruoxi Zhao